Worksheet: Demand and Elasticity

ECON 201

Definitions

Percentage change:

$$\%\text{ change} = \frac{\text{new} - \text{old}}{\text{old}} \times 100$$

Price elasticity of demand, ε: the percentage change in demand that would occur in response to a 1% increase in price,

$$\varepsilon = -\frac{\%\text{ change in } Q}{\%\text{ change in } P}$$

If price changes by ΔP and demand changes by ΔQ, three other equivalent ways to write it are

$$\varepsilon = -\frac{\Delta Q/Q}{\Delta P/P} = -\frac{P}{Q} \times \frac{\Delta Q}{\Delta P} = -\frac{P}{Q} \times \frac{1}{\text{slope}}$$

where the slope of the demand curve is $\Delta P/\Delta Q$.
Demand is **elastic** if ε is greater than 1, and **inelastic** if ε is less than 1.

In Lecture 3, you told us the most you would pay for a day at a Dodgers game. Linearizing your answers gives this demand curve:

$$Q = 90 - 0.3\,P$$

1. Elasticity from percentage changes

Use percentage changes to find the elasticity of demand for each $10 price rise below.

P rises from	Q falls from	% change in P	% change in Q	Elasticity, ε
$50 to $60				
$100 to $110				
$200 to $210				

At which prices is demand elastic?

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2. Elasticity from the formula

Use $\varepsilon = -\frac{P}{Q} \times \frac{\Delta Q}{\Delta P}$ to find the elasticity at the same three prices. With $Q = 90 - 0.3\,P$, $\Delta Q/\Delta P = -0.3$.

Price, P	$50	$100	$200
Demand, Q	75	60	30
Elasticity, ε			

Do you get the same answers as in Activity 1?
